Измерение и тестирование пропускной способности сети. Воспроизводимый эксперимент

Лабораторная работа № 3

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Цели и задачи

Основной целью работы является знакомство с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получение навыков проведения воспроизводимого эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

Выполнение лабораторной работы

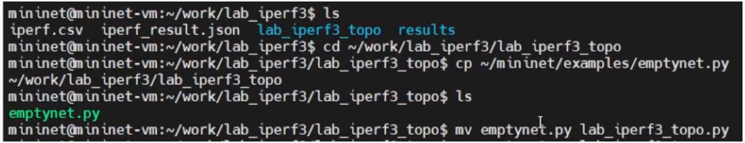
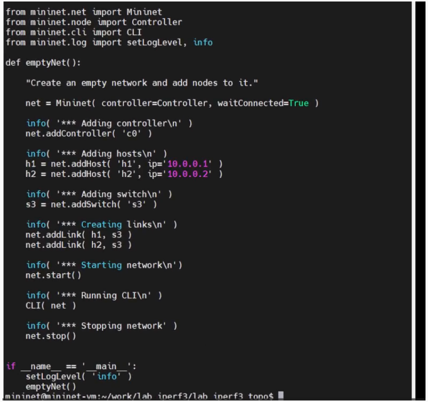


Рис. 1: создание необходимых каталогов

Создание топологии сети



```
from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3 )
    net.addLink( h2, s3 )

    info( '*** Starting network\n')
    net.start()

    info( '*** Running CLI\n' )
    CLI( net )

    info( '*** Stopping network' )
    net.stop()


if __name__ == '__main__':
    setLogLevel( 'info' )
    emptyNet()
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$
```

Рис. 2: скрипт

```
if __name__ == '__main__':
    setLogLevel( 'info' )
    emptyNet()
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_top
o.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
*** Running CLI
*** Starting CLI:
mininet> net
h1 h1-eth0:s3-eth1
h2 h2-eth0:s3-eth2
s3 lo:  s3-eth1:h1-eth0 s3-eth2:h2-eth0
c0
mininet> links
h1-eth0<->s3-eth1 (OK OK)
h2-eth0<->s3-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=928>
<Host h2: h2-eth0:10.0.0.2 pid=932>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=937>
<Controller c0: 127.0.0.1:6653 pid=921>
mininet> ex
```

Рис. 3: создание топологии сети

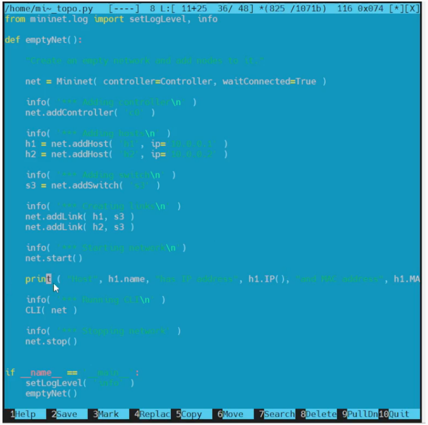


Рис. 4: внесениия изменениий в скрипт для вывода информации о хостах

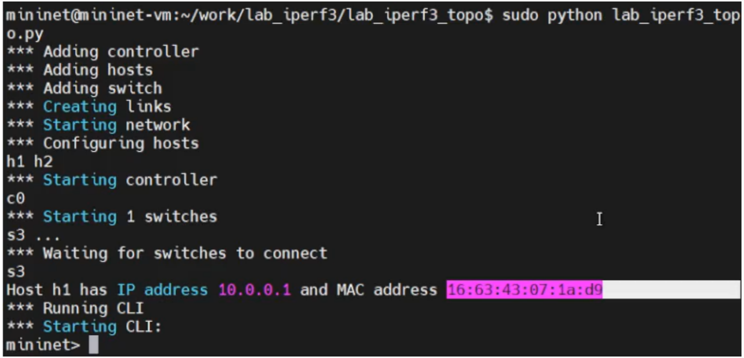


Рис. 5: топология сети

Создание топологии сети

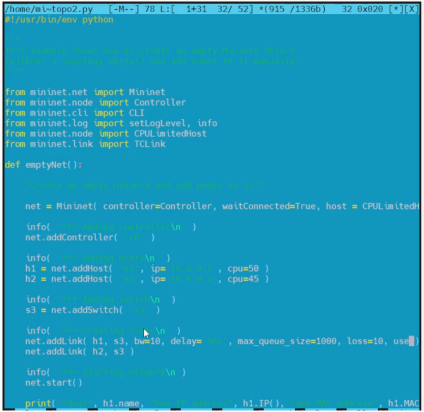
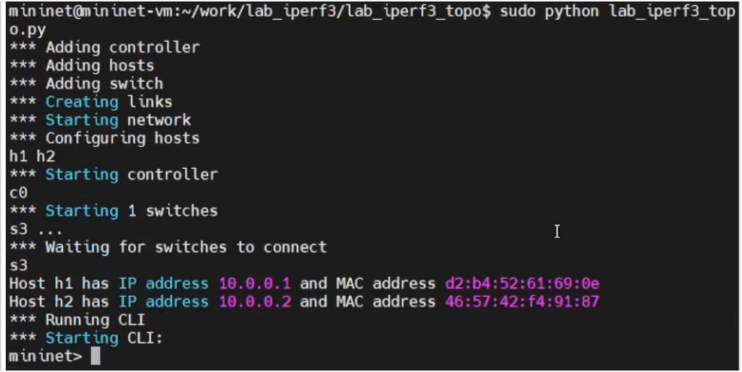


Рис. 6: внесение изменений в скрипт

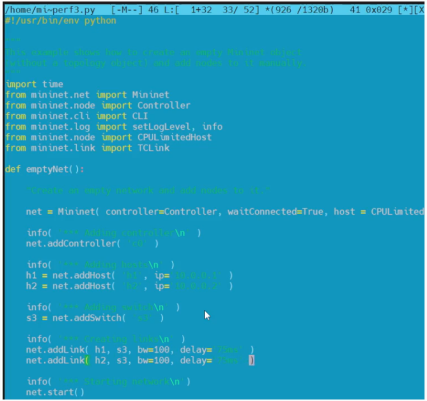
```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_top
o2.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(10.00Mbit 5ms delay 10.00000% loss) (10.00Mbit 5ms delay 10.00000% loss) *** St
arting network
*** Configuring hosts
h1 (cfs 5000000/100000us) h2 (cfs 4500000/100000us)
*** Starting controller
c0
*** Starting 1 switches
s3 (10.00Mbit 5ms delay 10.00000% loss) ...(10.00Mbit 5ms delay 10.00000% loss)
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address 36:bd:d6:74:51:f7
Host h2 has IP address 10.0.0.2 and MAC address 4e:25:b8:ae:64:ca
*** Running CLI
*** Starting CLI:
mininet>
```

Рис. 7: lab_iperf3_topo2.py



```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_top
o.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address d2:b4:52:61:69:0e
Host h2 has IP address 10.0.0.2 and MAC address 46:57:42:f4:91:87
*** Running CLI
*** Starting CLI:
mininet>
```

Рис. 8: lab_iperf3_topo.py



```
/home/mi~perf3.py   [-M--] 46 L:[  1+32  33/ 52] *(926 /1320b)   41 0x029 [*][X
#!/usr/bin/env python

"""
This example shows how to create an empty Mininet object
(without a topology object) and add nodes to it manually.
"""
import time
from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
from mininet.node import CPULimitedHost
from mininet.link import TCLink

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True, host = CPULimited

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3, bw=100, delay='75ms' )
    net.addLink( h2, s3, bw=100, delay='75ms' )

    info( '*** Starting network\n')
    net.start()
```

Рис. 9: добавление библиотек и изменение параметров на хостах

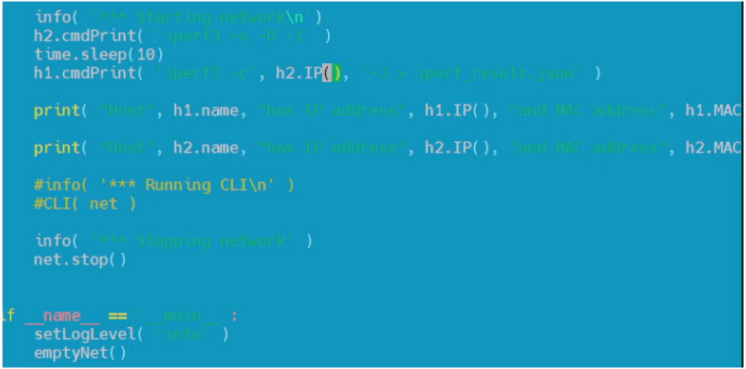


Рис. 10: внесение изменений в скрипт

Создание топологии сети

```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ sudo python lab_iperf3.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00M
bit 75ms delay) *** Starting network
*** Configuring hosts
h1 (cfs -1/100000us) h2 (cfs -1/100000us)
*** Starting controller
c0
*** Starting 1 switches
s3 (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) ...(100.00Mbit 75ms delay) (1
00.00Mbit 75ms delay)
*** Waiting for switches to connect
s3
*** Starting network
*** h2 : ('iperf3 -s -D -1',)
*** h1 : ('iperf3 -c', '10.0.0.2', '-J > iperf_result.json')
Host h1 has IP address 10.0.0.1 and MAC address d6:09:9b:02:95:c6
Host h2 has IP address 10.0.0.2 and MAC address 26:c8:e2:16:91:76
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
*** Done
```

Рис. 11: топология сети

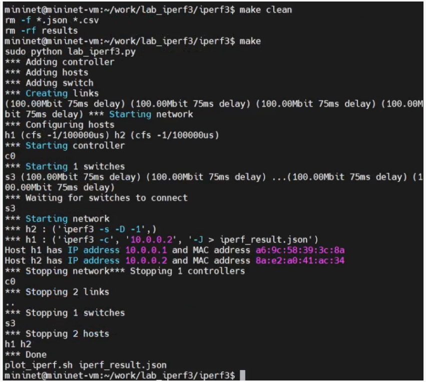
```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ plot_iperf.sh iperf_result.json
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ touch Makefile
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ ls
iperf.csv  iperf_result.json  lab_iperf3.py  Makefile  results
mininet@mininet-vm:~/work/lab_iperf3/iperf3$
```

Рис. 12: создание графиков и Makefile

Построение графиков эксперимента

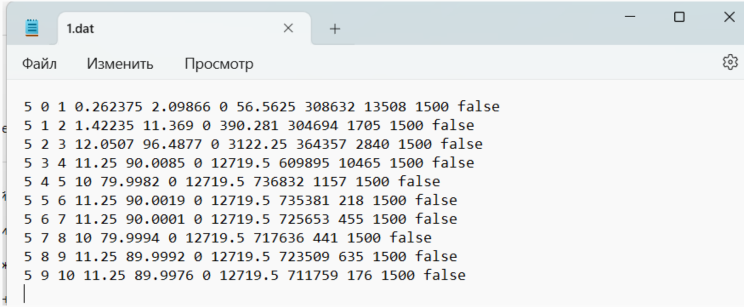
```
/home/mi~Makefile    [-M--] 23 L:[  1+10  11/ 11] *(178 / 178b) <EOF>
all: iperf_result.json plot

iperf_result.json:
<------->sudo python lab_iperf3.py

plot: iperf_result.json
<------->plot_iperf.sh iperf_result.json

clean:
<------->-rm -f *.json *.csv
<------->-rm -rf results
```

Рис. 13: Makefile



```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ make clean
rm -f *.json *.csv
rm -rf results
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ make
sudo python lab_iperf3.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00M
bit 75ms delay) *** Starting network
*** Configuring hosts
h1 (cfs -1/100000us) h2 (cfs -1/100000us)
*** Starting controller
c0
*** Starting 1 switches
s3 (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) ...(100.00Mbit 75ms delay) (1
00.00Mbit 75ms delay)
*** Waiting for switches to connect
s3
*** Starting network
*** h2 : ('iperf3 -s -D -1',)
*** h1 : ('iperf3 -c', '10.0.0.2', '-J > iperf_result.json')
Host h1 has IP address 10.0.0.1 and MAC address a6:9c:58:39:3c:8a
Host h2 has IP address 10.0.0.2 and MAC address 8a:e2:a0:41:ac:34
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
*** Done
plot_iperf.sh iperf_result.json
mininet@mininet-vm:~/work/lab_iperf3/iperf3$
```

Рис. 14: проверка отработки Makefile




```
5 0 1 0.262375 2.09866 0 56.5625 308632 13508 1500 false
5 1 2 1.42235 11.369 0 390.281 304694 1705 1500 false
5 2 3 12.0507 96.4877 0 3122.25 364357 2840 1500 false
5 3 4 11.25 90.0085 0 12719.5 609895 10465 1500 false
5 4 5 10 79.9982 0 12719.5 736832 1157 1500 false
5 5 6 11.25 90.0019 0 12719.5 735381 218 1500 false
5 6 7 11.25 90.0001 0 12719.5 725653 455 1500 false
5 7 8 10 79.9994 0 12719.5 717636 441 1500 false
5 8 9 11.25 89.9992 0 12719.5 723509 635 1500 false
5 9 10 11.25 89.9976 0 12719.5 711759 176 1500 false
```

Рис. 15: полученные графики

Построение графиков эксперимента

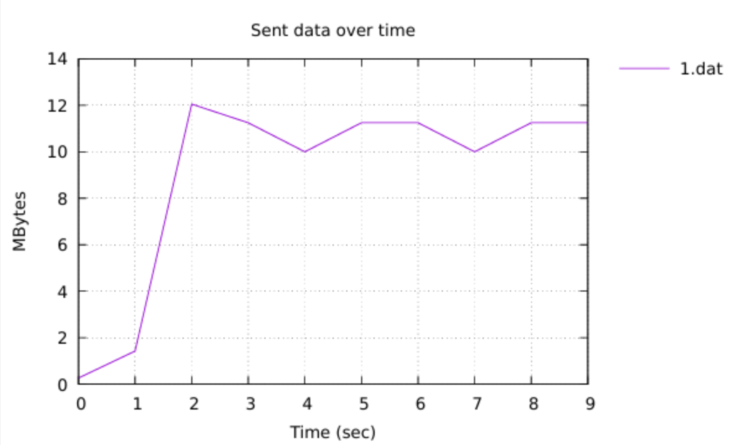


Рис. 16: количество переданных файлов

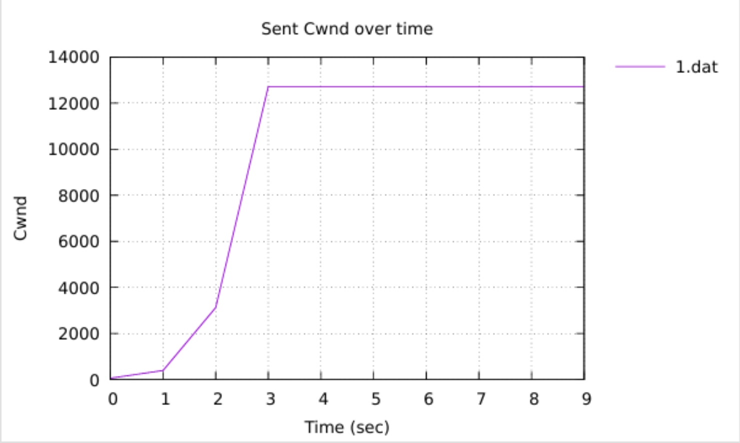


Рис. 17: окно перегрузки

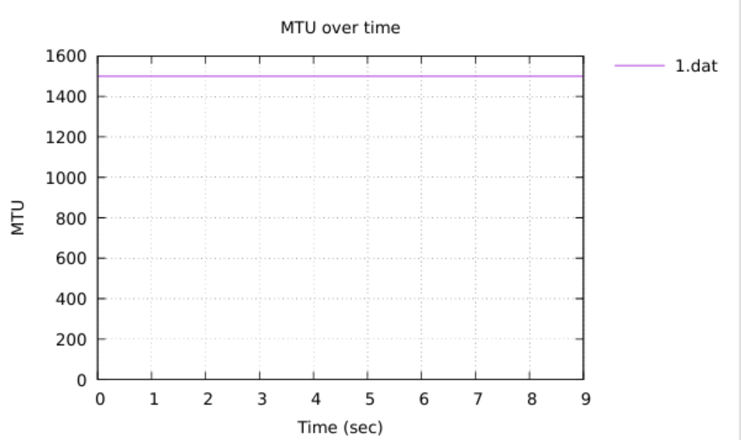


Рис. 18: максимальная единица передачи

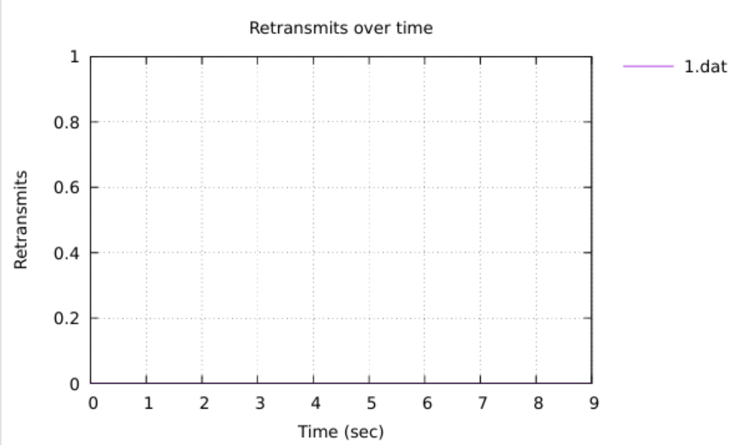


Рис. 19: повторная передача

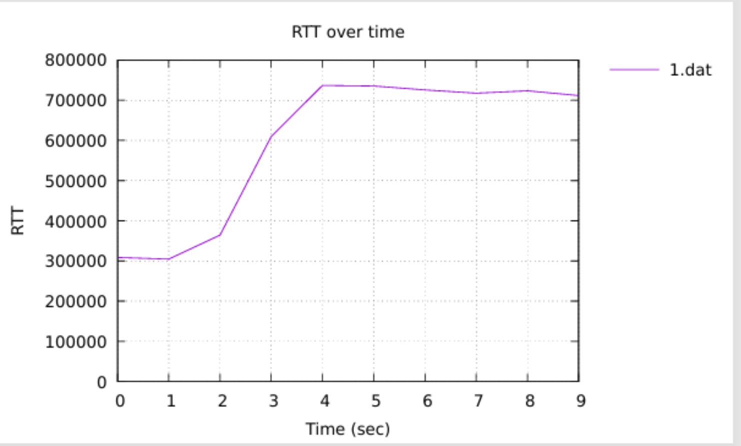


Рис. 20: время приема-передачи

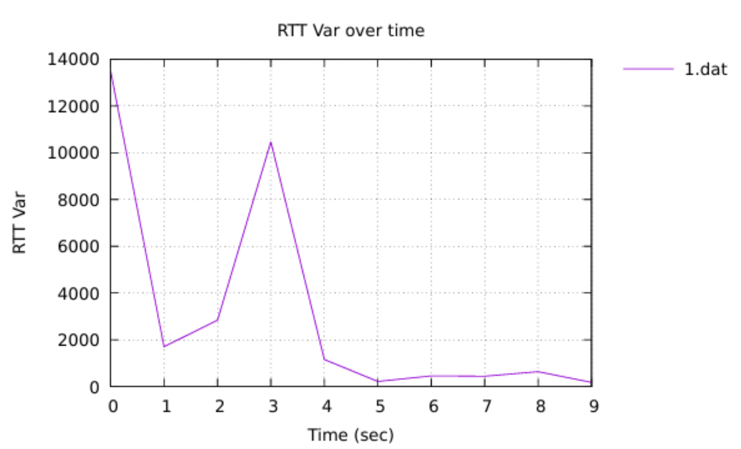


Рис. 21: отклонение времени приема-передачи

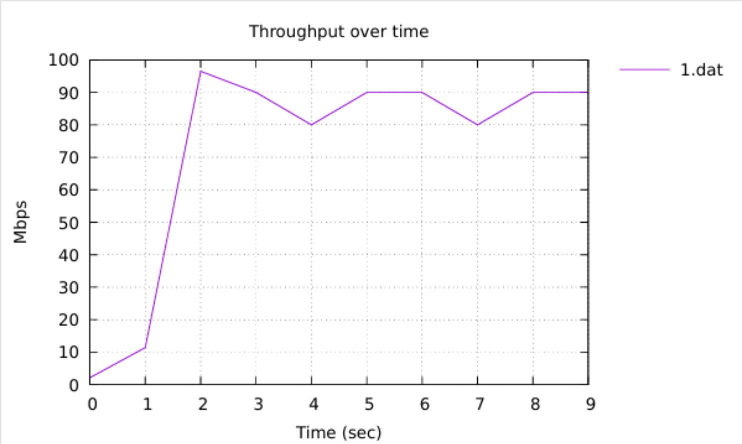


Рис. 22: пропускная способность

Выводы

Выводы

В результате выполнения лабораторной работы было проведено знакомство с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получение навыков проведения воспроизводимого эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.